

Natural Language Understanding (Spring 2026)

COURSE PROJECT INSTRUCTION

Project Objectives :

The objective of this project is to apply fine-tuned Large Language Models (LLMs) to accomplish specific tasks.

Project Discussion

Each team is required to prepare a presentation that includes the following components:

1. **Project Idea**
A clear and concise description of the task being addressed using a fine-tuned Large Language Model (LLM).
2. **Dataset Information**
Details about the dataset used, including source, size, structure, and any preprocessing steps performed.
3. **Base Model Information**
Information on the original (pre-trained) LLM used as the starting point, such as model name, architecture, and capabilities.
4. **Brief Explanation of Fine-Tuning with LoRA**
A high-level overview of Low-Rank Adaptation (LoRA) and how it was applied to fine-tune the base model.
5. **Evaluation Methods**
Metrics and techniques used to assess the model's performance, both before and after fine-tuning.
6. **Project Results and Plots**
A comparative analysis of the base model and the LoRA fine-tuned model using quantitative results and visual plots.
7. **Structured Output**
 - Showcase how the model produces structured, reliable, and task-specific outputs.

- Provide examples of outputs (e.g., in JSON format, tables, or other structured forms, depending on your task).

Note:

You must fine-tune the **selected base model** — not a different one. Ensure that the fine-tuning process, evaluation, and results are based on the same model you initially chose.

Project Ideas

1. Legal Document Summarization

Fine-tune an LLM to summarize long legal contracts or case law into short, readable summaries.

Dataset Example: Legal case documents or contracts from the Legal Case Reports dataset.

2. Invoice Information Extraction (IE)

Extract structured data (like invoice number, date, total, etc.) from scanned or text-based invoices.

Dataset Example: CORD Dataset or synthetic invoices.

3. Automated Feedback Generator for Student Essays

Generate personalized feedback based on student writing.

Dataset Example: Essays from Kaggle essay scoring datasets.

4. Clinical Note Classification or Diagnosis Suggestion

Classify or extract medical insights from clinical notes.

Dataset Example: MIMIC-III (with proper access).

5. Clinical Named Entity Recognition (NER)

Extract patient symptoms, diagnoses, medications, and procedures from clinical notes.

Dataset: MIMIC-III (with access), i2b2

6. Resume Parsing / Candidate Profiling

Extract structured info like name, education, skills, years of experience, etc., from resumes.

7. Opinion Mining from Customer Reviews

Extract sentiment and product features from reviews.

8. Product Search from Natural Queries

The goal is to build a system that allows users to search for products using natural language queries, rather than relying on keyword-based search alone. The challenge is to map the meaning of the user's query to the relevant products in the catalog, even if there is no exact match between the query and product descriptions.